Nama:Hafizh Arqamilandri Wakhyudi
NIM:103122400044

MODUL 3

TUGAS PENDAHULUAN

Aturan Tugas Pendahuluan :

1. Buatlah table menggunakan LiveSQL ataupun tools lainnya.
2. **Screenshoot** query dan hasil perubahan pada table.
3. Dilarang plagiarisme.

Soal Tugas Pedahuluan :

1. Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.
2. Lakukan insert data pada table nomor 1 sebanyak 15 rows.
3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:
   a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)
   b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH| YEAR]))
   c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))
   d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

JAWABAN

No.1

```
SQL> CREATE TABLE mahasiswa (
  2    id          NUMBER(3) PRIMARY KEY,
  3    nama        VARCHAR2(50) NOT NULL,
  4    tmp_lahir   VARCHAR2(30),
  5    tgl_lahir   DATE NOT NULL,
  6    no_hp       VARCHAR2(15),
  7    email       VARCHAR2(50),
  8    tinggi      NUMBER(5,2),
  9    berat       NUMBER(5,2)
```

10);

```
SQL> CREATE TABLE mahasiswa (
  2       id          NUMBER(3) PRIMARY KEY,
  3       nama        VARCHAR2(50) NOT NULL,
  4       tmp_lahir   VARCHAR2(30),
  5       tgl_lahir   DATE NOT NULL,
  6       no_hp       VARCHAR2(15),
  7       email       VARCHAR2(50),
  8       tinggi      NUMBER(5,2),
  9       berat       NUMBER(5,2)
 10  );

Table created.
```

```
SQL> desc mahasiswa;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ID                                        NOT NULL NUMBER(3)
 NAMA                                      NOT NULL VARCHAR2(50)
 TMP_LAHIR                                          VARCHAR2(30)
 TGL_LAHIR                                 NOT NULL DATE
 NO_HP                                              VARCHAR2(15)
 EMAIL                                              VARCHAR2(50)
 TINGGI                                             NUMBER(5,2)
 BERAT                                              NUMBER(5,2)

SQL>
```

No.2

```
SQL> INSERT ALL
  2     INTO mahasiswa VALUES (1, 'BUDI SANTOSO', 'Jakarta',
TO_DATE('12-05-2002','DD-MM-YYYY'), '081234567890', 'budi@email.com',
170.5, 65)
  3     INTO mahasiswa VALUES (2, 'ani wijaya', 'Bandung', TO_DATE('20-08-
2003','DD-MM-YYYY'), '085712345678', 'ani@email.com', 160.0, 50.5)
  4     INTO mahasiswa VALUES (3, 'Citra Lestari', 'Surabaya', TO_DATE('15-
01-2001','DD-MM-YYYY'), '081399887766', 'citra@email.com', 165.2, 55)
  5     INTO mahasiswa VALUES (4, 'dedi kurniawan', 'Medan', TO_DATE('05-
12-2002','DD-MM-YYYY'), '081122334455', 'dedi@email.com', 175.0, 80)
  6     INTO mahasiswa VALUES (5, 'Eka Putri', 'Jogja', TO_DATE('30-03-
2003','DD-MM-YYYY'), '082233445566', 'eka@email.com', 158.5, 48)
  7     INTO mahasiswa VALUES (6, 'Fajar Sidik', 'Semarang', TO_DATE('11-
07-2001','DD-MM-YYYY'), '089988776655', 'fajar@email.com', 180.0, 75.2)
  8     INTO mahasiswa VALUES (7, 'Gita Permata', 'Bali', TO_DATE('22-09-
2002','DD-MM-YYYY'), '087711223344', 'gita@email.com', 162.0, 52)
```

```
 9      INTO mahasiswa VALUES (8, 'HADI PRASETYO', 'Malang',
TO_DATE('09-02-2000','DD-MM-YYYY'), '081233445577', 'hadi@email.com',
168.0, 70)
 10     INTO mahasiswa VALUES (9, 'Indah Sari', 'Solo', TO_DATE('18-06-
2001','DD-MM-YYYY'), '085211223344', 'indah@email.com', 155.0, 45.8)
 11     INTO mahasiswa VALUES (10, 'Joko Susilo', 'Palembang',
TO_DATE('25-10-2002','DD-MM-YYYY'), '081311223344', 'joko@email.com',
172.5, 68)
 12     INTO mahasiswa VALUES (11, 'Kartika', 'Makassar', TO_DATE('03-04-
2003','DD-MM-YYYY'), '081199887744', 'kartika@email.com', 160.0, 58)
 13     INTO mahasiswa VALUES (12, 'Lutfi Hakim', 'Aceh', TO_DATE('14-11-
2001','DD-MM-YYYY'), '081288776655', 'lutfi@email.com', 178.0, 85)
 14     INTO mahasiswa VALUES (13, 'Maya Ahmad', 'Padang', TO_DATE('27-
08-2002','DD-MM-YYYY'), '085733445566', 'maya@email.com', 157.0, 50)
 15     INTO mahasiswa VALUES (14, 'Naufal', 'Balikpapan', TO_DATE('19-01-
2003','DD-MM-YYYY'), '081322334455', 'naufal@email.com', 170.0, 62.5)
 16     INTO mahasiswa VALUES (15, 'Olivia', 'Manado', TO_DATE('08-05-
2001','DD-MM-YYYY'), '082255443322', 'olivia@email.com', 163.4, 54)
 17  SELECT * FROM dual;
```

```
SQL> INSERT ALL
  2      INTO mahasiswa VALUES (1, 'BUDI SANTOSO', 'Jakarta', TO_DATE('12-05-2002','DD-MM-YYYY'), '081234567890', 'budi@email.com', 170.5, 65)
  3      INTO mahasiswa VALUES (2, 'ani wijaya', 'Bandung', TO_DATE('20-08-2003','DD-MM-YYYY'), '085712345678', 'ani@email.com', 160.0, 50.5)
  4      INTO mahasiswa VALUES (3, 'Citra Lestari', 'Surabaya', TO_DATE('15-01-2001','DD-MM-YYYY'), '081399887766', 'citra@email.com', 165.2, 55)
  5      INTO mahasiswa VALUES (4, 'dedi kurniawan', 'Medan', TO_DATE('05-12-2002','DD-MM-YYYY'), '081122334455', 'dedi@email.com', 175.0, 80)
  6      INTO mahasiswa VALUES (5, 'Eka Putri', 'Jogja', TO_DATE('30-03-2003','DD-MM-YYYY'), '082233445566', 'eka@email.com', 158.5, 48)
  7      INTO mahasiswa VALUES (6, 'Fajar Sidik', 'Semarang', TO_DATE('11-07-2001','DD-MM-YYYY'), '089988776655', 'fajar@email.com', 180.0, 75.2)
  8      INTO mahasiswa VALUES (7, 'Gita Permata', 'Bali', TO_DATE('22-09-2002','DD-MM-YYYY'), '087711223344', 'gita@email.com', 162.0, 52)
  9      INTO mahasiswa VALUES (8, 'HADI PRASETYO', 'Malang', TO_DATE('09-02-2000','DD-MM-YYYY'), '081233445577', 'hadi@email.com', 168.0, 70)
 10      INTO mahasiswa VALUES (9, 'Indah Sari', 'Solo', TO_DATE('18-06-2001','DD-MM-YYYY'), '085211223344', 'indah@email.com', 155.0, 45.8)
 11      INTO mahasiswa VALUES (10, 'Joko Susilo', 'Palembang', TO_DATE('25-10-2002','DD-MM-YYYY'), '081311223344', 'joko@email.com', 172.5, 68)
 12      INTO mahasiswa VALUES (11, 'Kartika', 'Makassar', TO_DATE('03-04-2003','DD-MM-YYYY'), '081199887744', 'kartika@email.com', 160.0, 58)
 13      INTO mahasiswa VALUES (12, 'Lutfi Hakim', 'Aceh', TO_DATE('14-11-2001','DD-MM-YYYY'), '081288776655', 'lutfi@email.com', 178.0, 85)
 14      INTO mahasiswa VALUES (13, 'Maya Ahmad', 'Padang', TO_DATE('27-08-2002','DD-MM-YYYY'), '085733445566', 'maya@email.com', 157.0, 50)
 15      INTO mahasiswa VALUES (14, 'Naufal', 'Balikpapan', TO_DATE('19-01-2003','DD-MM-YYYY'), '081322334455', 'naufal@email.com', 170.0, 62.5)
 16      INTO mahasiswa VALUES (15, 'Olivia', 'Manado', TO_DATE('08-05-2001','DD-MM-YYYY'), '082255443322', 'olivia@email.com', 163.4, 54)
 17  SELECT * FROM dual;

15 rows created.

SQL>
```

```
SQL> SELECT * FROM mahasiswa;

        ID NAME
---------- --------------------------------------------------
TMP_LAHIR                      TGL_LAHIR NO_HP
------------------------------ --------- ---------------
EMAIL                                                 TINGGI      BERAT
-------------------------------------------------- ---------- ----------
         1 BUDI SANTOSO
Jakarta                        12-MAY-02 081234567890
budi@email.com                                         170.5         65

         2 ani wijaya
Bandung                        20-AUG-03 085712345678
ani@email.com                                            160       50.5

        ID NAME
---------- --------------------------------------------------
TMP_LAHIR                      TGL_LAHIR NO_HP
------------------------------ --------- ---------------
EMAIL                                                 TINGGI      BERAT
-------------------------------------------------- ---------- ----------

         3 Citra Lestari
Surabaya                       15-JAN-01 081399887766
citra@email.com                                        165.2         55

         4 dedi kurniawan
Medan                          05-DEC-02 081122334455

        ID NAME
---------- --------------------------------------------------
TMP_LAHIR                      TGL_LAHIR NO_HP
------------------------------ --------- ---------------
```

No.3

A.Fungsi Karakter

```
SQL> SELECT
  2     UPPER(nama) as Kapital,
  3     LOWER(nama) as Kecil,
  4     INITCAP(nama) as Judul,
  5     LENGTH(nama) as Panjang_Nama,
  6     SUBSTR(nama, 1, 4) as Potong_4,
  7     REPLACE(email, 'email.com', 'itb.ac.id') as Ganti_Domain
  8  FROM mahasiswa WHERE id <= 5;
```

```
SQL> SELECT
  2     UPPER(nama) as Kapital,
  3     LOWER(nama) as Kecil,
  4     INITCAP(nama) as Judul,
  5     LENGTH(nama) as Panjang_Nama,
  6     SUBSTR(nama, 1, 4) as Potong_4,
  7     REPLACE(email, 'email.com', 'itb.ac.id') as Ganti_Domain
  8  FROM mahasiswa WHERE id <= 5;

KAPITAL
--------------------------------------------------
KECIL
--------------------------------------------------
JUDUL                                              PANJANG_NAME POTONG_4
-------------------------------------------------- ------------ ----------------
GANTI_DOMAIN
--------------------------------------------------------------------------------
BUDI SANTOSO
budi santoso
Budi Santoso                                                 12 BUDI
budi@itb.ac.id


KAPITAL
--------------------------------------------------
KECIL
--------------------------------------------------
JUDUL                                              PANJANG_NAME POTONG_4
-------------------------------------------------- ------------ ----------------
GANTI_DOMAIN
--------------------------------------------------------------------------------
ANI WIJAYA
ani wijaya
Ani Wijaya                                                   10 ani
ani@itb.ac.id
```

B.Fungsi Tanggal & Waktu

```
SQL> SELECT
  2     nama,
  3     TO_CHAR(tgl_lahir, 'DD MONTH YYYY') as Tgl_Lengkap,
  4     MONTHS_BETWEEN(SYSDATE, tgl_lahir)/12 as Umur_Tahun,
  5     LAST_DAY(tgl_lahir) as Akhir_Bulan_Lahir
  6  FROM mahasiswa WHERE id <= 5;
```

```
SQL> SELECT
  2      nama,
  3      TO_CHAR(tgl_lahir, 'DD MONTH YYYY') as Tgl_Lengkap,
  4      MONTHS_BETWEEN(SYSDATE, tgl_lahir)/12 as Umur_Tahun,
  5      LAST_DAY(tgl_lahir) as Akhir_Bulan_Lahir
  6  FROM mahasiswa WHERE id <= 5;

NAMA
--------------------------------------------------
TGL_LENGKAP                                UMUR_TAHUN AKHIR_BUL
------------------------------------------ ---------- ---------
BUDI SANTOSO
12 MAY       2002                          23.8329531 31-MAY-02

ani wijaya
20 AUGUST    2003                          22.5614478 31-AUG-03

Citra Lestari
15 JANUARY   2001                          25.1582219 31-JAN-01


NAMA
--------------------------------------------------
TGL_LENGKAP                                UMUR_TAHUN AKHIR_BUL
------------------------------------------ ---------- ---------
dedi kurniawan
05 DECEMBER  2002                           23.268437 31-DEC-02

Eka Putri
30 MARCH     2003                          22.9512327 31-MAR-03


SQL>
```

C.Fungsi Angka (tinggi,berat)

SQL> SELECT
2 nama,
3 ROUND(tinggi) as Tinggi_Bulat,
4 CEIL(berat) as Berat_Keatas,
5 MOD(id, 2) as Cek_Ganjil_Genap
6 FROM mahasiswa WHERE id <= 5;

```
SQL> SELECT
  2      nama,
  3      ROUND(tinggi) as Tinggi_Bulat,
  4      CEIL(berat) as Berat_Keatas,
  5      MOD(id, 2) as Cek_Ganjil_Genap
  6  FROM mahasiswa WHERE id <= 5;

NAMA                                           TINGGI_BULAT BERAT_KEATAS
---------------------------------------------- ------------ ------------
CEK_GANJIL_GENAP
----------------
BUDI SANTOSO                                            171           65
               1

ani wijaya                                              160           51
               0

Citra Lestari                                           165           55
               1


NAMA                                           TINGGI_BULAT BERAT_KEATAS
---------------------------------------------- ------------ ------------
CEK_GANJIL_GENAP
----------------
dedi kurniawan                                          175           80
               0

Eka Putri                                               159           48
               1


SQL>
```

D.Fungsi Konversi

```
SQL> SELECT
  2     nama,
  3     CONCAT(TO_CHAR(tinggi, '999.9'), ' cm') AS Tinggi_Teks,
  4     TO_CHAR(tgl_lahir, 'Day') AS Hari_Lahir
  5  FROM mahasiswa
  6  WHERE id <= 5;
```

```
SQL> SELECT
  2     nama,
  3     CONCAT(TO_CHAR(tinggi, '999.9'), ' cm') AS Tinggi_Teks,
  4     TO_CHAR(tgl_lahir, 'Day') AS Hari_Lahir
  5  FROM mahasiswa
  6  WHERE id <= 5;

NAMA                                               TINGGI_TE
-------------------------------------------------- ---------
HARI_LAHIR
-----------------------------------
BUDI SANTOSO                                        170.5 cm
Sunday

ani wijaya                                          160.0 cm
Wednesday

Citra Lestari                                       165.2 cm
Monday


NAMA                                               TINGGI_TE
-------------------------------------------------- ---------
HARI_LAHIR
-----------------------------------
dedi kurniawan                                      175.0 cm
Thursday

Eka Putri                                           158.5 cm
Sunday


SQL>
```

Nama:Hafizh Arqamilandri Wakhyudi SE-08-02
NIM:103122400044

Semangat!!